

# Biology concepts

BioInf 2025/2026 1<sup>st</sup> Semester

André Lamúrias

[a.lamurias@fct.unl.pt](mailto:a.lamurias@fct.unl.pt)

# Summary

- Overview of key biomolecules
- Biological processes
- Genome

# Recap

- Biological data is sequential (DNA, RNA, proteins)
- Bioinformatics uses computational approaches to study these data
- To develop these approach, it is necessary to understand some biology basic concepts
- (Helps to talk “biology” with other bioinformaticians)

# Starting small

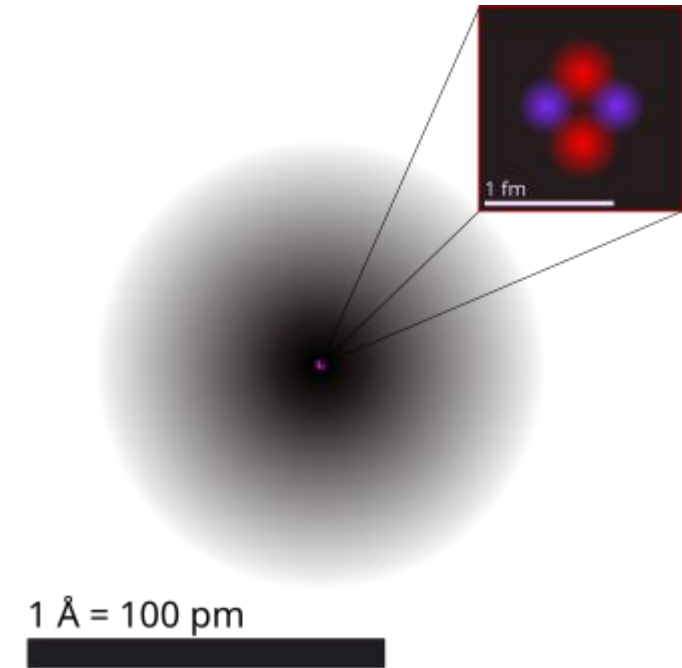
- Atoms

The periodic table is color-coded by the origin of element synthesis. A legend at the top identifies the following categories:

- Big Bang fusion:** Light blue (H, He)
- Cosmic ray fission:** Red (Fr, Ra)
- Dying low-mass stars:** Green (Li, Be, B, C, N, O, F, Ne, Si, P, S, Cl, Ar, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, As, Se, Br, Kr, Rb, Sr, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, I, Xe, Cs, Ba, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, At, Rn, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr)
- Exploding massive stars:** Yellow (Na, Mg, Al, Si, P, S, Cl, Ar, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, As, Se, Br, Kr, Rb, Sr, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, I, Xe, Cs, Ba, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, At, Rn, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr)
- Human synthesis No stable isotopes:** Grey (Tc, Pm, At, Rn, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr)
- Merging neutron stars:** Purple (Fr, Ra, Cs, Ba, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, At, Rn, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr)
- Exploding white dwarfs:** Light grey (Fe, Co, Ni, Cu, Zn, Ga, Ge, As, Se, Br, Kr, Rb, Sr, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, I, Xe, Cs, Ba, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, At, Rn, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr)

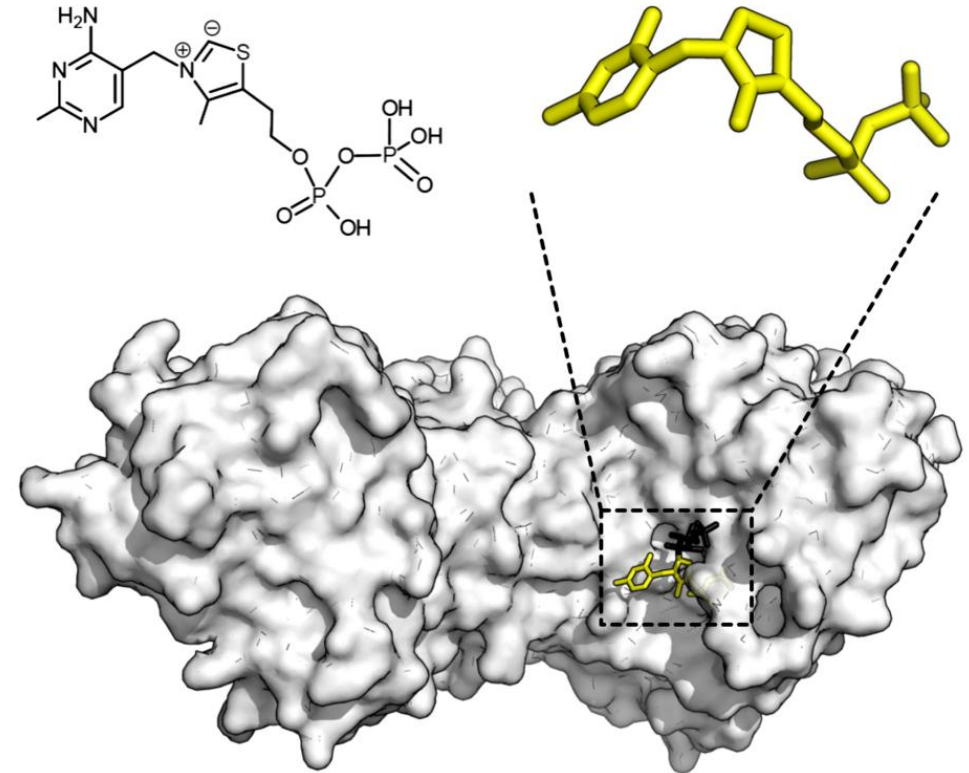
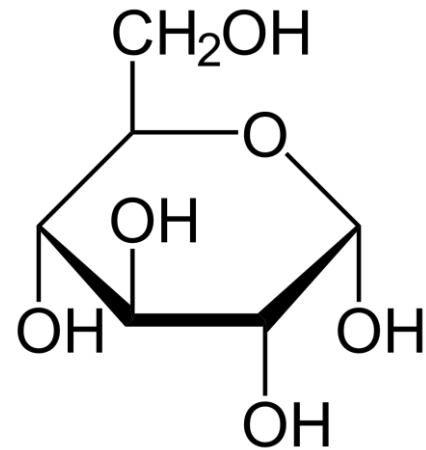
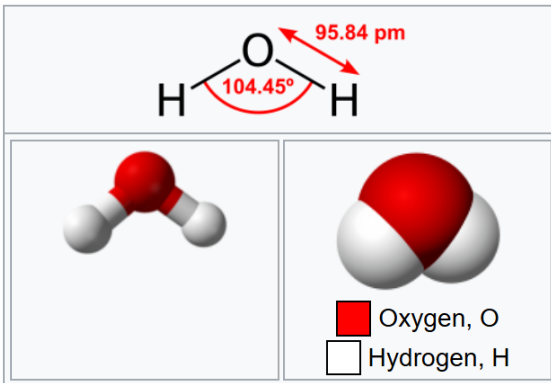
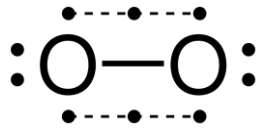
Elements are highlighted with circles: H (1), C (6), N (7), O (8), P (15), and S (16).

Periodic Table



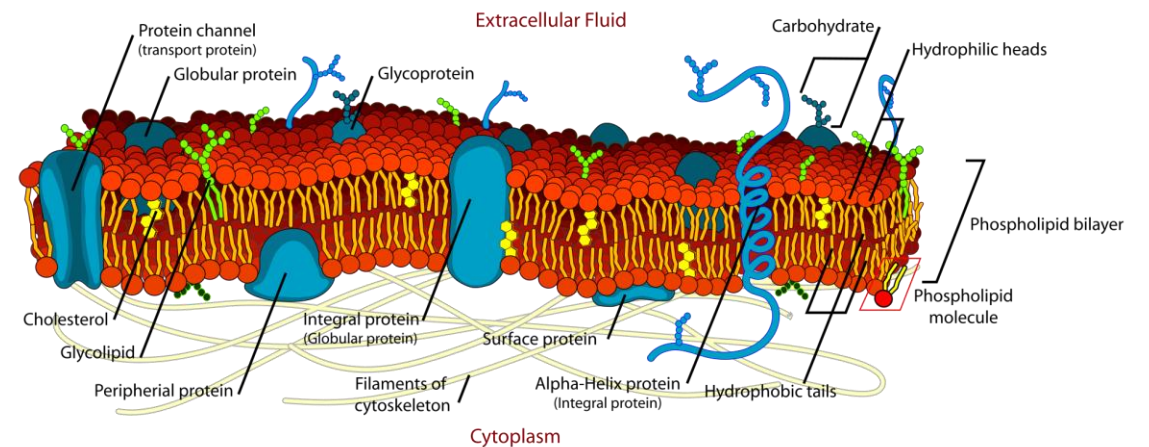
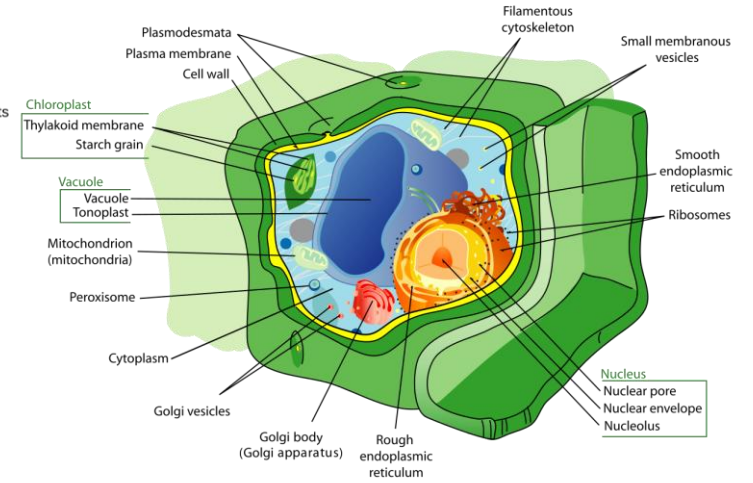
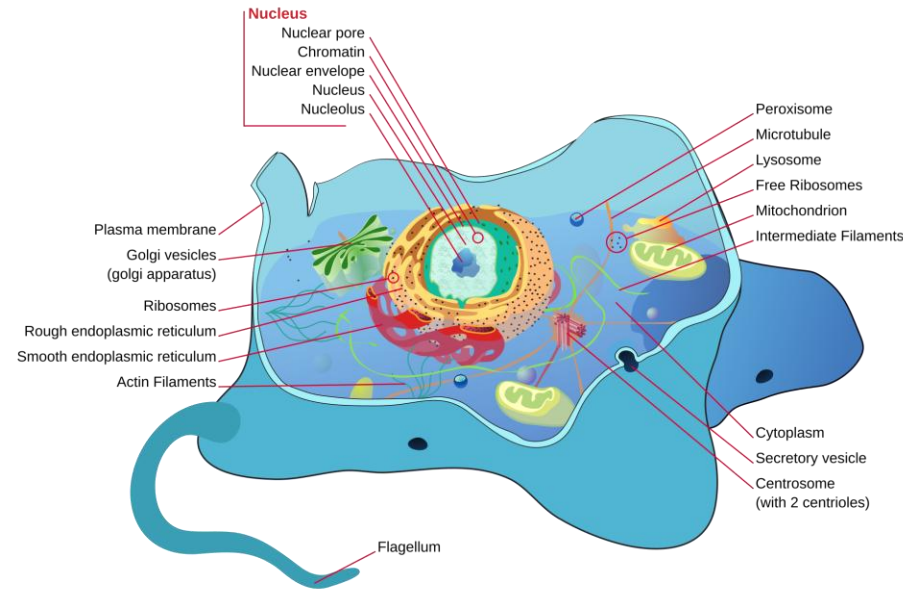
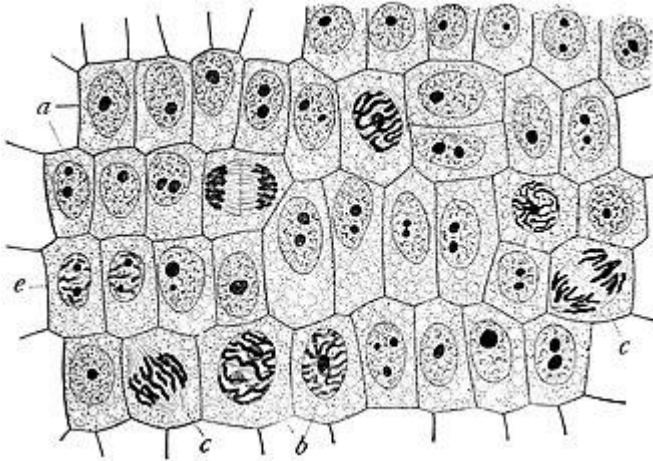
# Starting small

- Molecules



# Starting small

- Cells

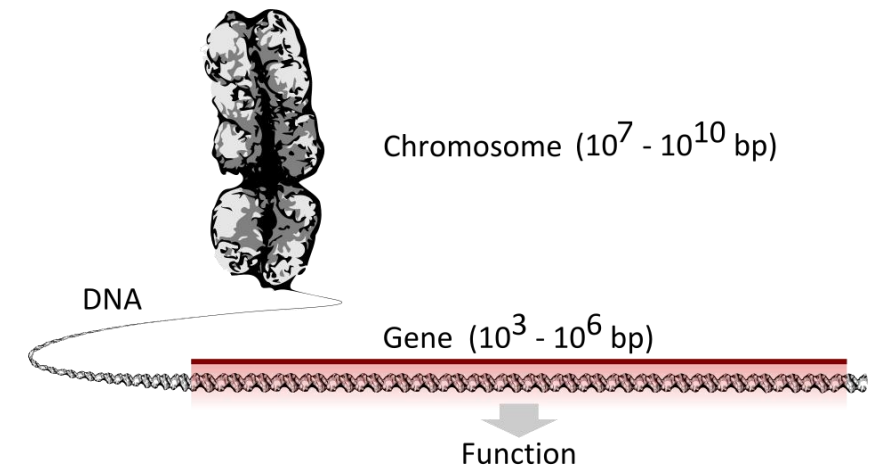
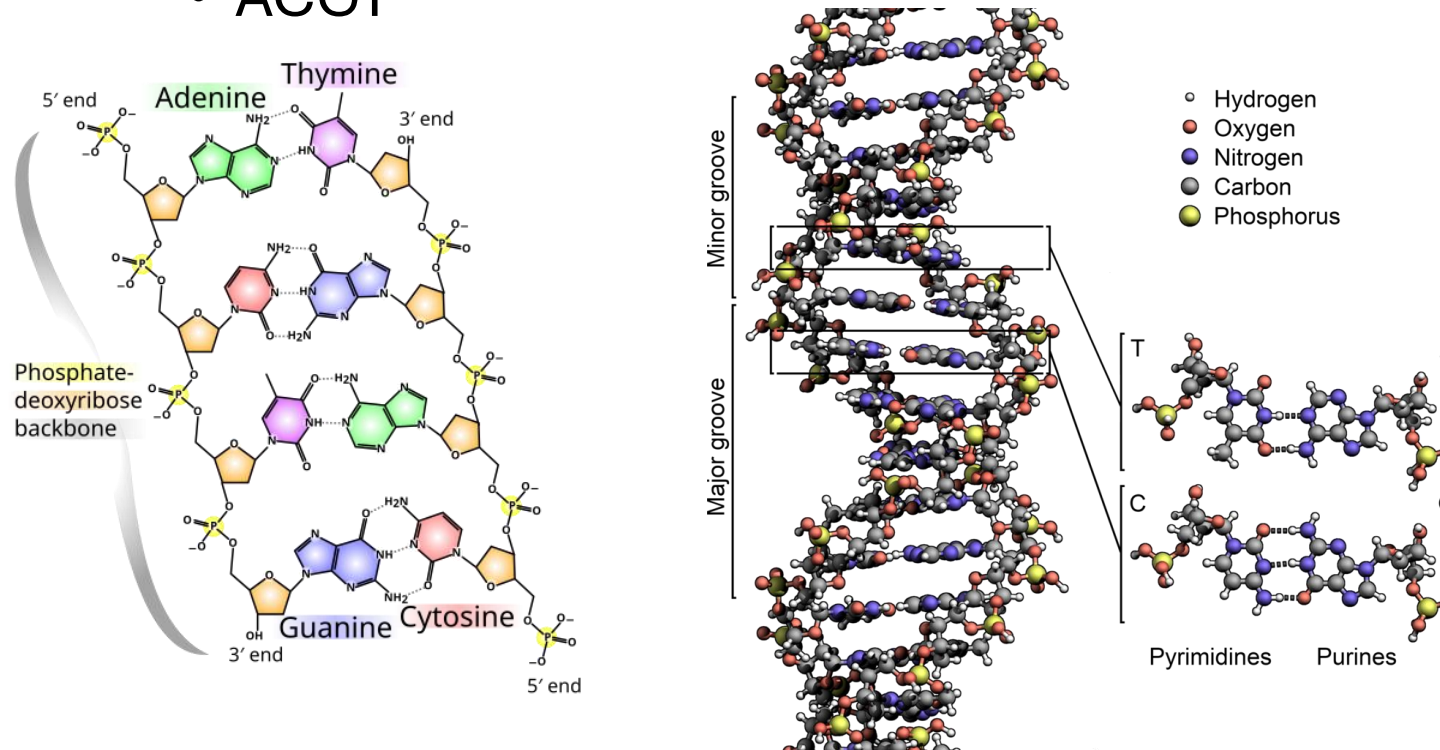


# Biomolecules



# Biomolecules - DNA

- DNA: Deoxyribonucleic acid
  - ACGT



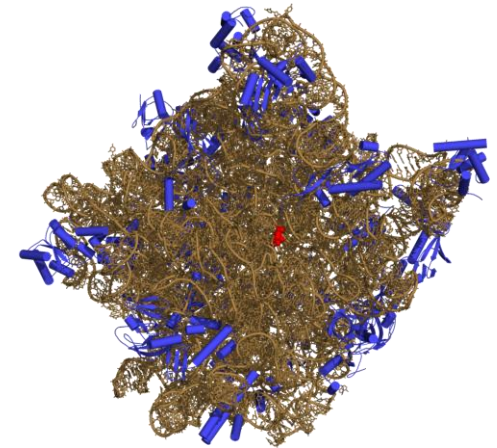
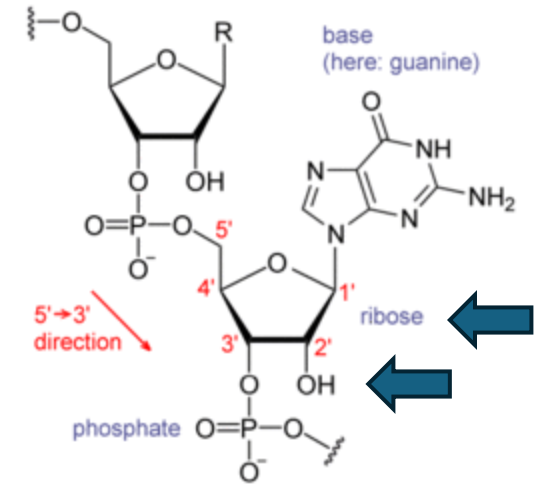


# Biomolecules - DNA

- Purine nucleotides: **A**denine and **G**uanine
- Pyrimidine nucleotides: **C**ytosine and **T**hymine
- A-T; G-C
- Nucleotides = Bases
- Complementary DNA strands – Base pairs (bp)
- DNA strand has two ends: 5' and 3'
- Where is it in the cell:
  - Animals, plants, fungi: nucleus of the cell, mitochondria, chloroplasts
  - Bacteria: cytoplasm

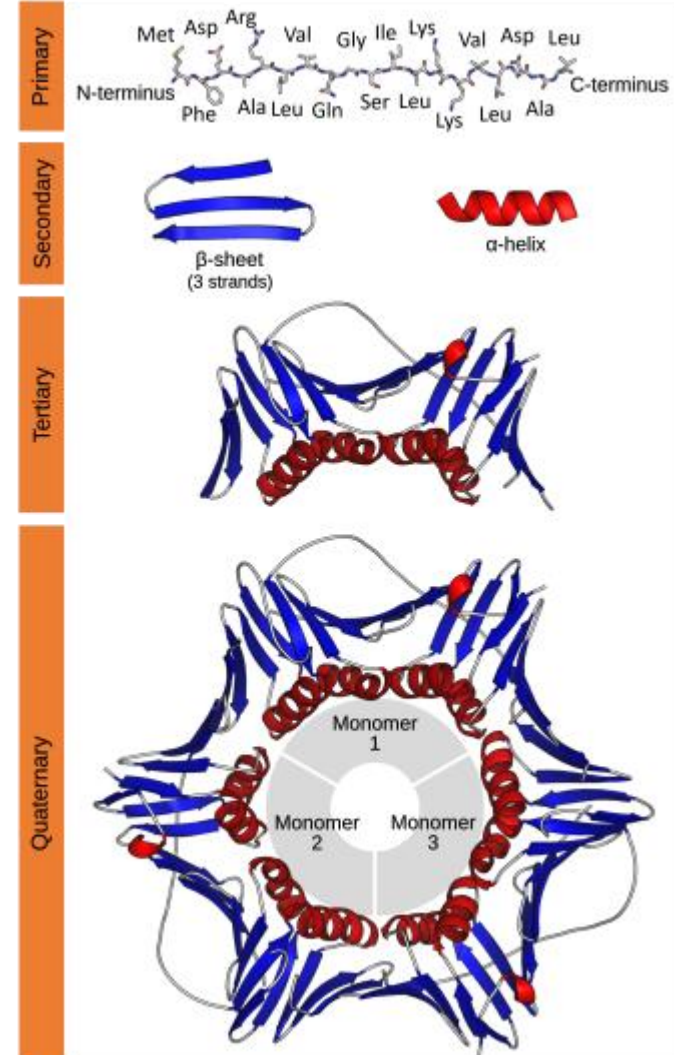
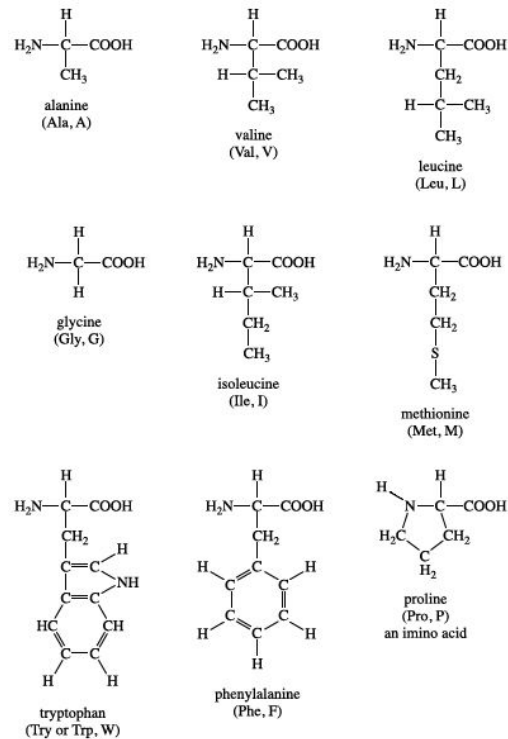
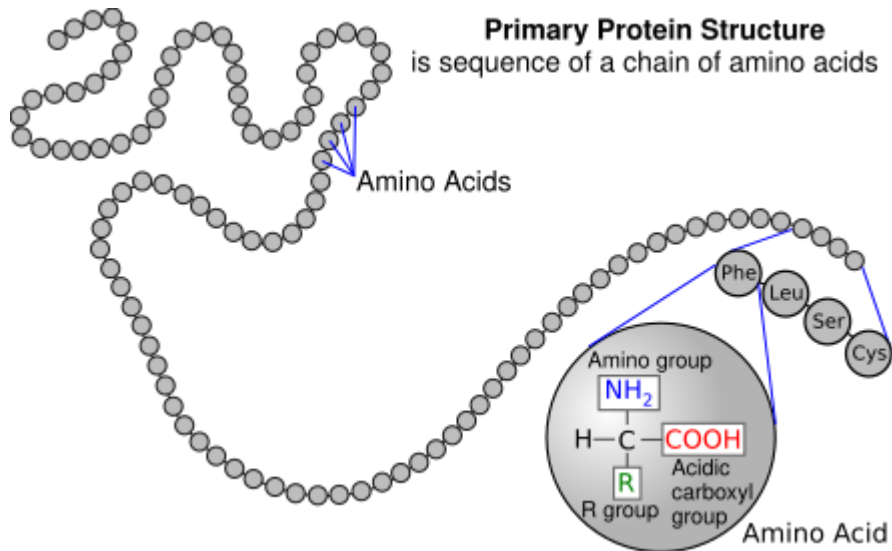
# Biomolecules - RNA

- RNA: Ribonucleic acid
  - ACGU (A-U, C-G)
- Synthesized using DNA as the template (transcription)
- Several types:
  - Messenger RNA (mRNA): gene expression
  - Transfer RNA (tRNA): carry aminoacids for protein synthesis
  - Ribosomal RNA (rRNA): produce chains of aminoacids
- Usually single stranded, various conformations
- In the past, only RNA was used, no DNA



# Biomolecules - Proteins

- Chains of aminoacids (if short-length: Peptides)



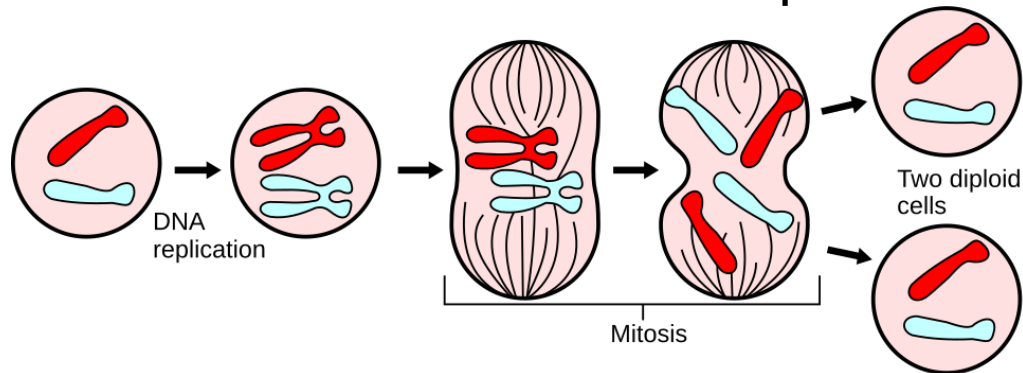
# Biomolecules - Proteins

- Only about 20 aminoacids are used in proteins
- Various functions in a cell:
  - Metabolic reactions (enzymes) e.g. transform sugar into energy
  - DNA replication
  - Cell structure
  - Transport molecules
- Fundamental question: can we predict the function of a protein based on its sequence of aminoacids?

# Biological processes

# Biological processes - DNA replication

- DNA contains “genetic information” (genome)
- Cells multiply and need to preserve DNA molecule (sequence of nucleotides)
- If we change the sequence (mutations), it can have an effect on the new cell
  - We want the exact same sequence



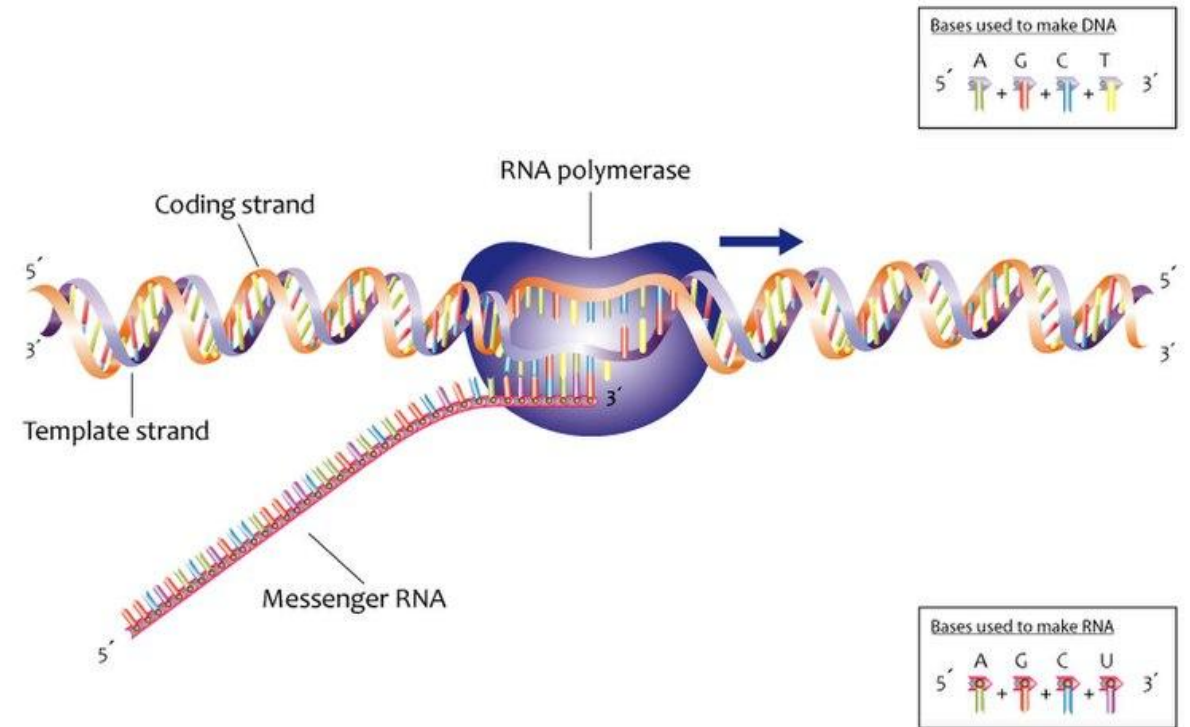
# Biological processes - Transcription

- DNA contains “genetic information” (genome)
  - Determines which proteins can be produced
- To produce proteins from DNA, we first need to produce mRNA – **transcription**
- More mRNA -> more protein
- mRNA does not cover the whole chromosome
  - Transcription starts and ends at specific places in a DNA sequence
  - Protein may need several RNAs to be produced



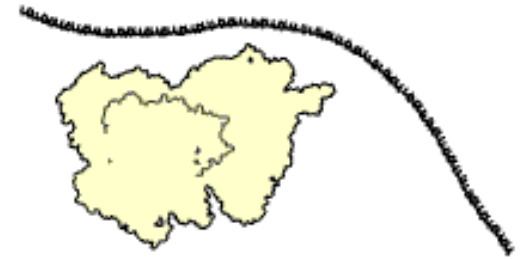
# Biological processes - Transcription

- Only one of the DNA strands is transcribed
  - One strand is called the template strand (RNA will be complementary to this sequence (A→U, T→A, G→C, C→G))
  - The other strand is called the coding strand (RNA will be like this sequence except T→U)

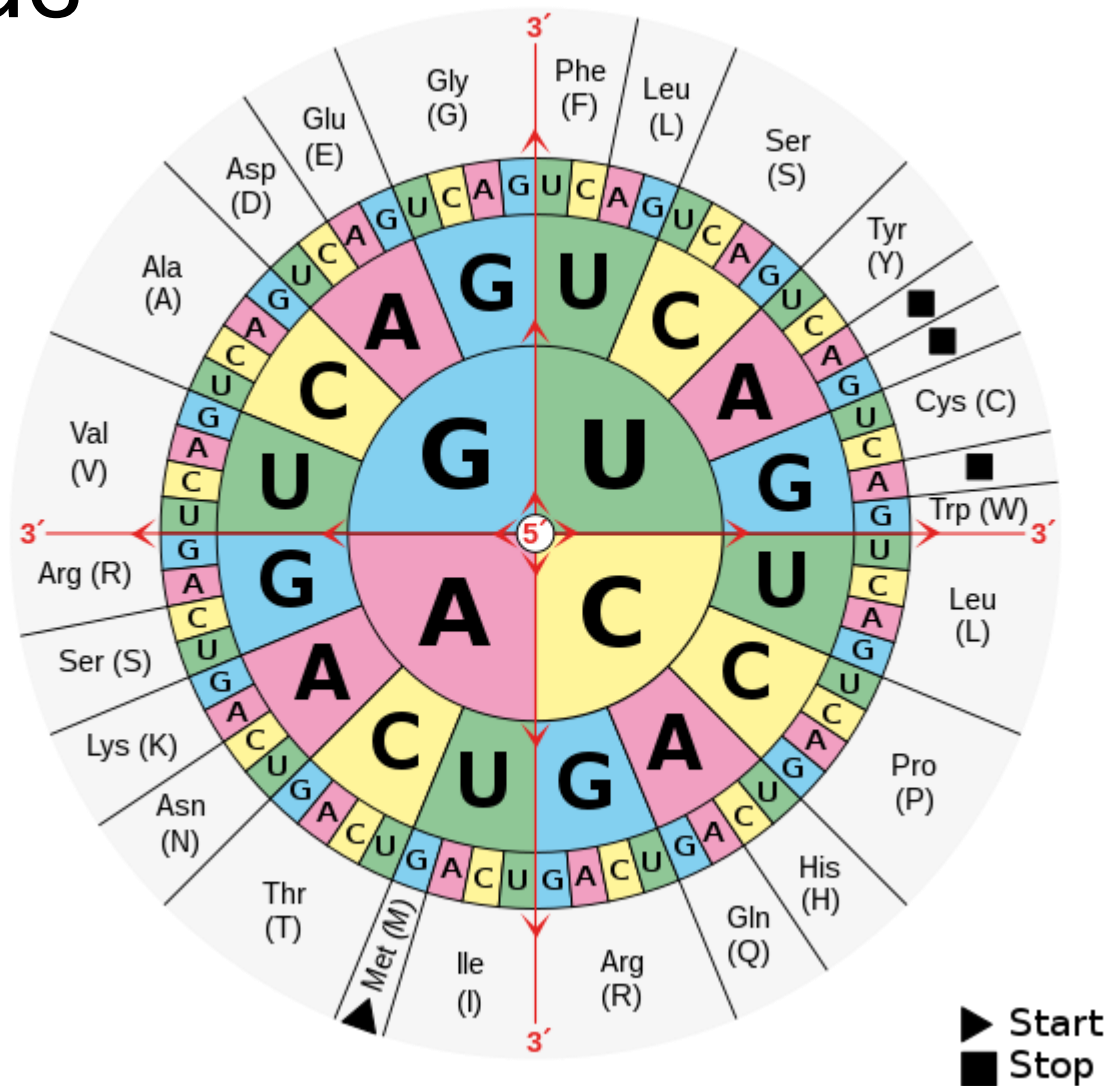


# Biological processes - Translation

- Synthesize proteins using mRNA
- Ribosomes read mRNA in non-overlapping groups of 3 nucleotides -> **codons**
- Happens outside the nucleus
- Each codon corresponds to a single aminoacid: genetic code
  - How many different codons can exist?



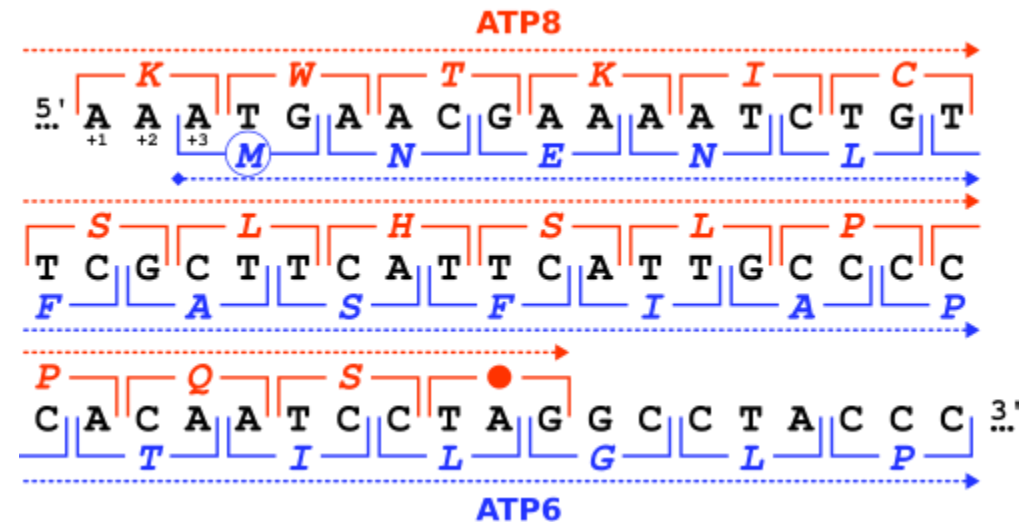
# Genetic code



# Genetic code

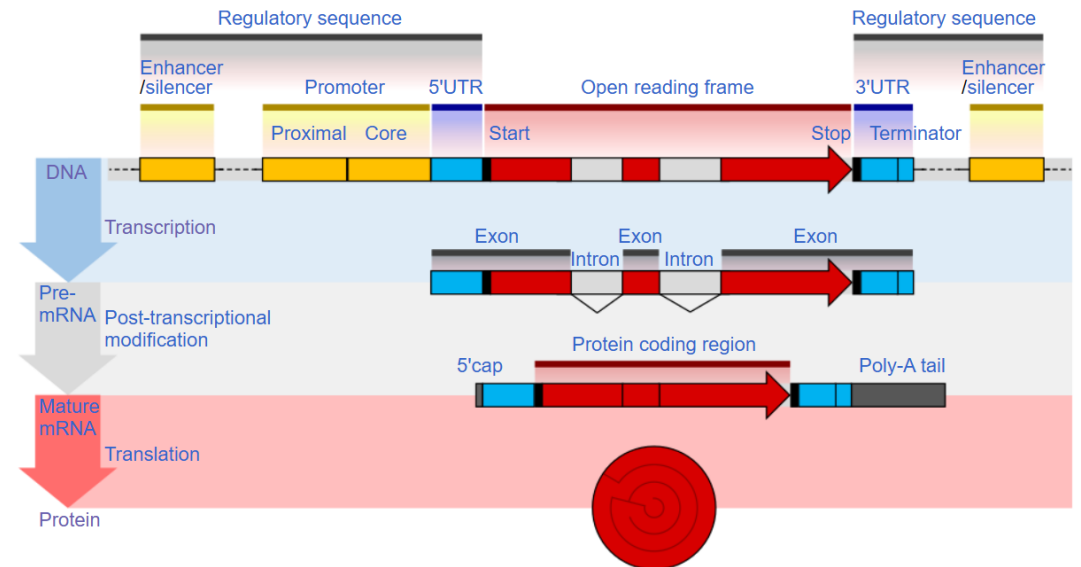
- Depending on the position we start, we can have different codons, leading to a different protein – reading frames

| AAUGCUCGUAUUUUAG                           |
|--------------------------------------------|
| AAU-GCU-CGU-AAU-UUA → Asn-Ala-Arg-Asn-Leu  |
| AUG-CUC-GUA-AUU-UAG → Met-Leu-Val-Ile-Stop |
| UGC-UCG-UAA-UUU → Cys-Ser-Stop-Stop        |

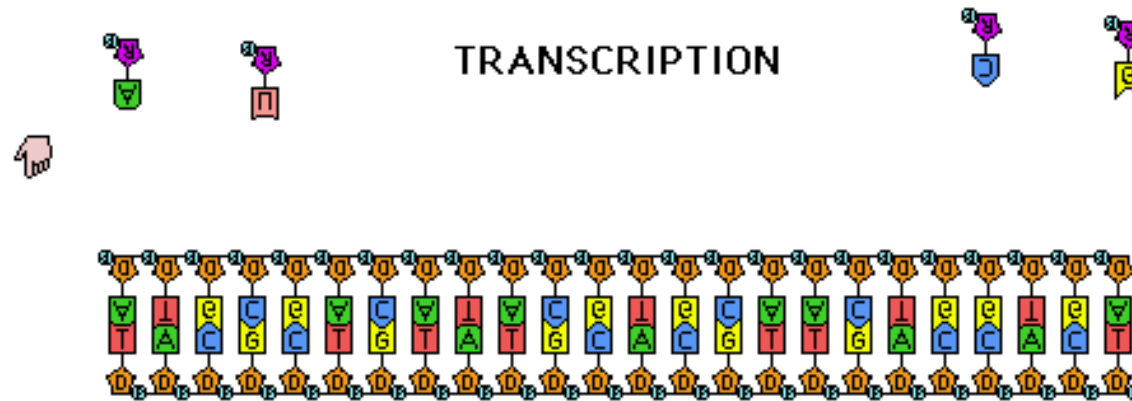


# Genes

- Basic unit of hereditary
- Functional regions of the DNA and encode the instructions to make a protein
- Genome: Collection of all genetic material
- Genotype: set of genes of an organism
- Phenotype: characteristics of an organism determined by its genes



# Translation and Transcription



(in nucleus)

# Conclusion

- DNA, RNA, Proteins are the fundamental biomolecules, each with distinct structures and roles.
- Information flows from DNA → RNA → Protein (Central Dogma)
- Replication, Transcription, Translation are the core processes.
  - These processes generate the sequential data (strings!) that we will analyze computationally.

# Further reading

- Rocha et al. 2018 Chapter 3
- [Biology for Computer Scientists](#) -- Lawrence Hunter